

## UNIT 6



# DATA REPRESENTATION AND LOGIC GATES



## DATA REPRESENTATION IN A COMPUTER

- Digital devices use electronic circuits that work with two states: 0 and 1.
- These circuits are built from logic gates, which perform basic logical operations.
- Data is stored, processed, and transmitted in this binary format.

## LOGIC GATES

Logic gates are the fundamental building blocks of electronic circuits in digital systems, carrying out logical operations such as AND, OR, NOT, NAND, and NOR. These gates are categorized into two groups: basic and universal logic gates.

## BASIC LOGIC GATES

Logic gates serve as the foundational elements of digital logic circuits, with the primary ones being AND, OR, and NOT gates. These gates execute their respective AND, OR, and NOT logical functions within a digital system.

## AND GATE

An AND gate is a digital circuit that has two or more inputs and a single output. AND gate operates on logical multiplication rules. AND operation using variables A and B is represented as "A.B", where (.) is a logical multiplication sign.

Boolean Expression of AND gate:  $Y = A.B$

**Truth Table of AND operation using two input variables:**

| Truth table of and operation using two input variables |   |           |
|--------------------------------------------------------|---|-----------|
| A                                                      | B | $Y = A.B$ |
| 0                                                      | 0 | 0         |
| 0                                                      | 1 | 0         |
| 1                                                      | 0 | 0         |
| 1                                                      | 1 | 1         |



Fig. 6.2  
AND Gate using two input variables

Truth Table of AND gate using two input variables A, B, and output is Y. If any input is 0, then output Y becomes 0. If all inputs are 1, then output Y becomes 1.

| Truth table of and operation using three input variables |   |   |             |
|----------------------------------------------------------|---|---|-------------|
| A                                                        | B | C | $Y = A.B.C$ |
| 0                                                        | 0 | 0 | 0           |
| 0                                                        | 0 | 1 | 0           |
| 0                                                        | 1 | 0 | 0           |
| 0                                                        | 1 | 1 | 0           |
| 1                                                        | 0 | 0 | 0           |
| 1                                                        | 0 | 1 | 0           |
| 1                                                        | 1 | 0 | 0           |
| 1                                                        | 1 | 1 | 1           |

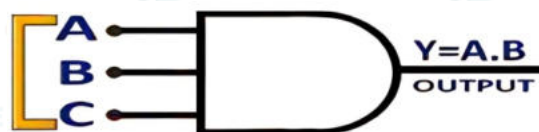


Fig. 6.3  
AND Gate using three input variables

AND gate using three input variables A, B, C, and output is Y. If any input is 0, then output Y becomes 0. If all inputs are 1, then output Y becomes 1.

## OR GATE

An OR gate is a type of digital logic circuit characterized by multiple input lines and a single output line. It functions according to the principles of logical addition. When using variables A and B, the logical OR operation is denoted as "A+B," where the symbol (+) signifies logical addition. The Boolean expression for an OR gate's output, Y, is given by  $Y = A + B$ .

| Truth table of or gate operation using two input variables |   |         |
|------------------------------------------------------------|---|---------|
| A                                                          | B | $Y=A+B$ |
| 0                                                          | 0 | 0       |
| 0                                                          | 1 | 1       |
| 1                                                          | 0 | 1       |
| 1                                                          | 1 | 1       |

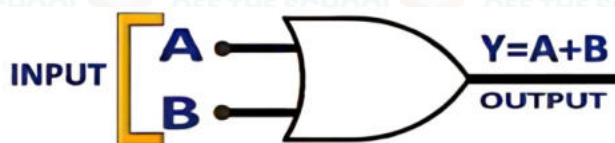


Fig. 6.4 OR Gate using two input variables

Truth Table of OR gate using two input variables A, B, and Y is output. If any input is 1, then output Y becomes 1, and if all inputs are 0, then output Y becomes 0. Boolean expression of OR gate is  $Y = A + B$ .

| Truth table of or operation using three input variables |   |   |           |
|---------------------------------------------------------|---|---|-----------|
| A                                                       | B | C | $Y=A+B+C$ |
| 0                                                       | 0 | 0 | 0         |
| 0                                                       | 0 | 1 | 1         |
| 0                                                       | 1 | 0 | 1         |
| 0                                                       | 1 | 1 | 1         |
| 1                                                       | 0 | 0 | 1         |
| 1                                                       | 0 | 1 | 1         |
| 1                                                       | 1 | 0 | 1         |
| 1                                                       | 1 | 1 | 1         |

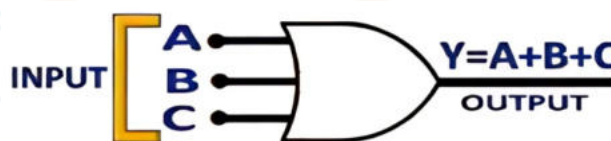


Fig. 6.5 OR Gate using three input variables

Truth Table of OR gate using three input variables A, B, C and Y is output. If any input is 1, then output Y becomes 1, and if all inputs are 0, then output Y becomes 0.

## NOT GATE

A NOT gate is a digital circuit that has a single input and a single output. It is also known as an INVERTER. The output of the NOT gate is the logical inversion of input. It is symbolically represented by the complement sign (') on the right side of the input variable or a bar (-) sign on top of the variable.

Boolean Expression of NOT gate:  $Y = A'$  or  $Y = \bar{A}$

**Truth table of NOT gate operation using two input variables:**

| Truth table of not gate operation using two input variables |               |
|-------------------------------------------------------------|---------------|
| A                                                           | $Y = \bar{A}$ |
| 0                                                           | 1             |
| 1                                                           | 0             |



Fig. 6.6 NOT Gate

Truth table of NOT gate is A as input and  $Y = A'$  is output.

## UNIVERSAL GATES

A universal gate is a logic gate which can implement any Boolean function without the need to use any other type of basic gates. The NOR gate and NAND gate are universal gates.

## NAND (NOT - AND) GATE

The NAND gate or "Not AND" gate is the collection of two basic logic gates, the AND gate and the NOT gate connected in series.

Boolean Expression of NAND gate:  $Y = (A.B)'$  or  $Y = \overline{A.B}$

| Truth table of nand operation using two input variables |   |                            |
|---------------------------------------------------------|---|----------------------------|
| A                                                       | B | $Y = \overline{A \cdot B}$ |
| 0                                                       | 0 | 1                          |
| 0                                                       | 1 | 1                          |
| 1                                                       | 0 | 1                          |
| 1                                                       | 1 | 0                          |

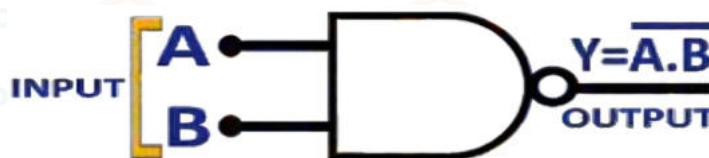


Fig. 6.8 NAND Gate

The Truth table of the NAND gate using two input variables A, B and Y is the output. When all inputs are "1", the output Y is "0". If any one of the inputs is "0", then the output Y is "1".

## NOR GATE

A NOR Gate is the collection of OR Gate NOT Gate. The output of NOR gate is inverter OR. Boolean Expression of NOR gate:  $Y = (A + B)'$  or  $Y = \overline{A + B}$ .

| Truth table of nor operation using two input variables |   |                        |
|--------------------------------------------------------|---|------------------------|
| A                                                      | B | $Y = \overline{A + B}$ |
| 0                                                      | 0 | 1                      |
| 0                                                      | 1 | 0                      |
| 1                                                      | 0 | 0                      |
| 1                                                      | 1 | 0                      |

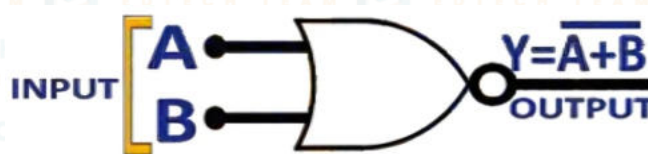


Fig. 6.9 NOR Gate

The Truth table of the NOR gate using two input variables A, B and Y is the output. If both inputs are "0", then the output Y is "1". If any one of the inputs is "1", then the output Y is "0".

## Difference Between Basic Logic Gates And Universal Logic Gates:

| Basic Logic Gates                                                                             | Universal Logic Gates                                                                                      |
|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| The most basic logic gates are AND, OR, and NOT.                                              | NAND and NOR gates are considered universal gates because they can be used to create any other logic gate. |
| By using these three gates, it is possible to implement all of the basic Boolean expressions. | A universal gate is a gate which can implement any Boolean expression without using other types of gate.   |

## BOOLEAN ALGEBRA

Boolean algebra, created by George Boole in 1954, is a mathematical field that provides a framework for analyzing logical operations and the manipulation of binary data. It operates on binary inputs—true or false—to generate a corresponding output of true or false.

## RULES OF BOOLEAN ALGEBRA

The principles of Boolean algebra are encapsulated in a series of predefined rules, known as the Laws of Boolean Algebra Expressions, which are designed to streamline logical expressions. These 12 foundational rules facilitate the simplification of logical gate arrangements, reducing the complexity and number of gates required to execute a specific logical function.

## BOOLEAN ALGEBRA RULES AND PROOFS

| Rule        | Proof                                                 |
|-------------|-------------------------------------------------------|
| $A + 0 = A$ | Identity law: Adding zero leaves the value unchanged. |

|                          |                                                                              |
|--------------------------|------------------------------------------------------------------------------|
| $A + 0 = A$              | Identity law: Adding zero leaves the value unchanged.                        |
| $A \cdot 1 = A$          | Identity law: Multiplying by one leaves the value unchanged.                 |
| $A + 1 = 1$              | Domination law: Any value OR true (1) results in true (1).                   |
| $A \cdot 0 = 0$          | Domination law: Any value AND false (0) results in false (0).                |
| $A + A = A$              | Idempotent law: ORing a value with itself does not change it.                |
| $A \cdot A = A$          | Idempotent law: ANDing a value with itself does not change it.               |
| $A + A' = 1$             | Complement law: A value ORed with its complement results in true (1).        |
| $A \cdot A' = 0$         | Complement law: A value ANDed with its complement results in false (0).      |
| $A'' = A$                | Double negation law: Negating a value twice returns to the original value.   |
| $A + A \cdot B = A$      | Absorption law: Simplifies to A, showing the irrelevance of the second term. |
| $A + A' \cdot B = A + B$ | Absorption law: Combines the product of the complement with OR.              |